Predicting Human Motion in Human–Robot Collaboration Using PINNs and Computer Vision

A Physics-Informed Neural Network Approach That Combines Vision-Based Human Tracking With Human Biomechanics

1. The Problem We're Solving

Today, collaborative robots react to what a human has already done. A camera sees a hand move, the system processes it, and only then does the robot respond. That delay — however small — is the difference between a smooth handover and a jerky, machine-like reaction. It's also a safety gap: if the robot could know a fraction of a second earlier where a human hand is heading, it could begin a safe, corrective motion before contact ever becomes a risk.

The goal of this project is narrow and deliberate: predict where the human body (specifically the arm/hand) will be a short time into the future — roughly 100–500 milliseconds ahead — during human-robot collaboration tasks. We are not building the robot's response. We are building the anticipation layer that a robot's controller could later use.

2. Why Not Just Use Reinforcement Learning or Plain Deep Learning?

It's tempting to reach for a big data-driven model, but two common approaches fall short here:

- Reinforcement Learning (RL) learns a policy that maps a situation to an action, guided by a reward signal. Our problem has no "action" or "reward" — we are only forecasting where a human body will be next, not deciding what a robot should do about it. RL is also very data-hungry, and struggles to generalize to human movements it hasn't seen before — which is exactly the situation you cannot afford in a safety-critical, human-facing environment.
- Plain deep learning (e.g. an LSTM or Transformer trained only on trajectories) can genuinely predict motion, but with nothing forcing its output to obey the laws of physics or human biomechanics, it can produce trajectories that are statistically plausible but physically impossible — a wrist accelerating faster than a human joint can move, or a jump in position when the camera briefly loses track of the hand.

This is why the project centers on Physics-Informed Neural Networks (PINNs): models that are trained on real data but are also constrained, mathematically, to respect the physical/biomechanical rules of how a human body can actually move. This keeps predictions smooth, realistic, and far more reliable under noisy or incomplete vision data than a purely data-driven model.

3. Our Approach: Two Technologies, One Pipeline

> *Use computer vision to see the human, and human biomechanics equations (built into a PINN) to constrain what the network is allowed to predict about where that human will move next.*

This project rests on two technologies working together, not one replacing the other:

Technology	What it's responsible for	Why it's needed
Computer Vision (CV)	Watches the human through video and estimates their current joint positions/angles, frame by frame — this is the model's "eyes."	Without CV, the model has no idea where the human currently is. But CV alone can only report coordinates — it has no understanding of what the human body is physically capable of doing next.
Physics-Informed Neural Network (PINN)	Takes that stream of CV-estimated positions and predicts the next 100–500ms of motion,	Without physics constraints, a network trained on noisy CV data can predict

Technology	What it's responsible for	Why it's needed
	while being mathematically constrained by human biomechanics equations — this is the model's "understanding."	trajectories that are statistically plausible but physically impossible (sudden jumps, unrealistic accelerations).

Neither piece alone solves the problem. CV supplies the raw perception; the PINN's biomechanics constraint supplies the anticipation. Together, the system doesn't just see the human — it anticipates them, and stays reliable even when the camera's view is imperfect.

At training time, the model is shown both the noisy CV-estimated trajectory and a clean IMU-based trajectory (used as ground truth, and to compute the biomechanics constraint). At deployment time, only a camera is needed — no motion-capture suit — because the model has learned to produce physically consistent predictions even from imperfect visual input alone.

4. Dataset: InHARD

After comparing two candidate datasets, InHARD (Industrial Human Action Recognition Dataset) was chosen as the primary and only dataset for this project. The alternative, ACTIVE, is built for long-range (3–50m), moving-camera robot perception and is better suited to whole-body action recognition than to precise, joint-level motion prediction — so it was set aside.

Property	Details
Scale	2,000,000+ frames, 4,800+ action samples, 16 subjects
Action classes	13 industrial actions (assembling parts, using tools, passing items to a cobot arm)
Skeleton data	17 body joints, 3D position + rotation, captured at 120Hz via an IMU motion suit (Perception Neuron 32), stored as BVH files
RGB data	3 synchronized camera views (top, left-45°, right-45°) per action
Robot involved	UR10 collaborative arm — some actions directly involve human–robot interaction

The dual-modality nature of InHARD is what makes it a strong fit: the IMU skeleton is precise enough to serve as ground truth and to compute biomechanics quantities, while the synchronized RGB video lets us simulate exactly what a real, camera-only deployed robot would actually see.

5. How the Vision Pipeline Extracts 3D Motion

Concretely, turning InHARD's video into the joint-trajectory input the PINN needs works one of two ways:

- Multi-view triangulation (preferred): InHARD provides 3 synchronized camera views (top, left-45°, right-45°) of the same action. If camera calibration parameters are available, 2D keypoints detected in each view can be triangulated into accurate 3D joint positions — avoiding the depth ambiguity a single camera has.
- Monocular 3D pose lifting (fallback): if calibration data isn't available, a single-view 3D pose estimation model is used instead. This is noisier — but it's also the more realistic real-world case for a robot with just one camera, and exactly where the PINN's physics constraint earns its keep, smoothing over depth/occlusion errors that CV alone can't resolve.

Either way, the output of this step is the same: a per-frame estimate of joint positions/angles, which becomes the PINN's input signal — the same role the IMU skeleton plays during training as ground truth.

6. What "Human Biomechanics Equations" Actually Means Here

It's important to be precise about the level of biomechanics this project uses, so the scope is honest and defensible. There are three levels of increasing detail:

Level	What it models	Data required	Used in this project?
Joint kinematics	Positions/angles only, no forces	What InHARD gives directly	Baseline input, not the constraint
Joint dynamics (rigid-body biomechanics)	Net torque at each joint (shoulder, elbow, wrist)	Kinematics + standard anthropometric segment tables (mass, length, inertia) — computable, no extra data collection needed	Yes — this is the physics constraint used in the PINN
Muscle-level dynamics (Hill-type models)	Individual muscle forces/activations	EMG data, or an offline OpenSim optimization pipeline — not available in InHARD	Not in current scope — noted as future work

In short: this project uses rigid multi-body joint dynamics — the standard physics used across human movement science and ergonomics — not individual muscle modeling. That is enough to keep predictions physically realistic while staying achievable within the project's scope.

7. How the Physics Is Built Into the Model

The prediction network is trained with a combined loss function — not just "match the data," but "match the data AND never violate physics":

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L = λ1·L_data + λ2·L_dynamics + λ3·L_smoothness + λ4·L_limits
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Term	What it enforces
L_data	Predicted future joint positions/velocities should match the real observed continuation of the motion
L_dynamics	The predicted trajectory should satisfy $M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau$ — i.e., be consistent with real joint-torque dynamics, computed from anthropometric body-segment parameters
L_smoothness	Human reaching motion is empirically close to "minimum jerk" — penalizes unnaturally jerky predicted paths
L_limits	Penalizes predictions that exceed realistic joint angle, velocity, or torque limits of the human arm

Where q represents joint angles (from the skeleton data), and M, C, G are standard rigid-body dynamics terms built from anthropometric tables (e.g., de Leva, 1996) scaled to each subject.

8. Scoping Decision: Upper Body First

Nearly all of InHARD's relevant actions — reaching, handover, assembling parts, passing items to the robot — are upper-body dominant. Modeling the full 17-joint body adds significant complexity without adding much value for this use case.

- Phase 1 (recommended starting scope): shoulder – elbow – wrist chain only (a 3-joint arm model). Fully covers handover, reaching, and proximity-safety scenarios.
- Phase 2 (optional, time permitting): extend to full-body dynamics for more general industrial actions.

9. What This Project Deliberately Does Not Include

- Robot response/control — the project stops at predicting human motion; it does not decide what the robot should do with that prediction.
- Muscle-level force/activation modeling — noted as a possible future extension via an OpenSim-based pipeline, not part of the current deliverable.
- Reinforcement learning — not a natural fit for a forecasting problem with no reward signal; the real baseline comparison is against purely data-driven sequence models (see below).

10. How We'll Know It Works

The PINN will be compared against a purely data-driven baseline (e.g., an LSTM or Transformer trained on the same trajectories, but without the physics loss terms), across:

- Prediction accuracy: Mean Per-Joint Position Error (MPJPE) at multiple horizons — 100ms, 200ms, 500ms ahead.
- Physical plausibility: bone-length consistency over time, and how often predictions violate realistic joint limits.
- Data efficiency: train both models on shrinking fractions of the dataset (100%, 50%, 10%) — the PINN is expected to degrade more gracefully, since physics fills in what limited data cannot teach.
- Robustness to occlusion: mask parts of the input trajectory (simulating a hand briefly hidden from camera view) and compare how gracefully each model handles the gap.

11. End-to-End Pipeline Summary

- Step 1 — Data: InHARD RGB video (noisy input) + IMU skeleton (clean supervision target), synchronized.
- Step 2 — Vision: extract 3D joint trajectories from RGB video (triangulation across camera views, or monocular 3D pose estimation).
- Step 3 — Physics setup: compute anthropometric body-segment parameters and joint-torque dynamics from the IMU skeleton data via inverse dynamics.
- Step 4 — Model: train a PINN that takes a short window of observed joint trajectory and predicts the next 100–500ms, with a loss combining data-fit and physics-consistency terms.
- Step 5 — Evaluation: compare against a non-physics-informed baseline on accuracy, physical plausibility, data efficiency, and occlusion robustness.

12. Why This Matters

If the system knows the physics of how a human arm moves, it can anticipate where that hand will be a couple hundred milliseconds ahead. That means a robot could begin its own corrective motion at the same time as the human, rather than after. It turns human-robot interaction from a series of reactive, jerky corrections into something closer to a synchronized partnership — a robot that feels like it's actually paying attention, not just processing frames after the fact.